ISEBE LEMFUNDO LEMPUMA KOLONI
EASTERN CAPE EDUCATION DEPARTMENT
OOS-KAAP ONDERWYSDEPARTEMENT

NATIONAL SENIOR CERTIFICATE

GRADE 11

ENGINEERING GRAPHICS AND DESIGN P2

NOVEMBER 2016

EXAMINATION

MARKS: 200

TIME: 3 hours

This question paper consists of 6 pages.

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INSTRUCTIONS AND INFORMATION

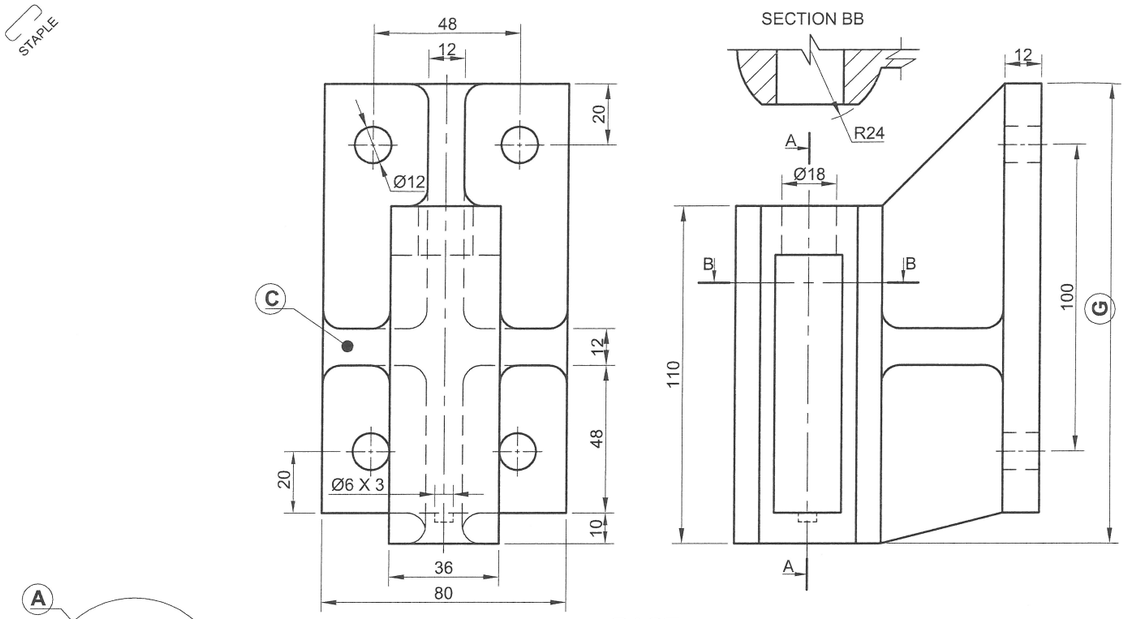
1. The paper consists of FOUR questions.
2. Answer ALL the questions.
3. All drawings must be drawn to scale 1:1, unless otherwise stated.
4. All questions must be answered on the answer sheets provided.
5. All the answers sheets must be re-stapled in numerical sequence and handed in irrespective of whether the question was attempted or not.
6. Careful time management is essential in order to complete all the questions.
7. Print your name in the block provided on every answer sheet.
8. All answers must be drawn accurately and neatly.
9. Any details or dimensions not given must be estimated in good proportion.

FOR OFFICIAL USE ONLY					
				MODERATED MARK	
1					
2					
3					
4					
TOTAL					
	2	0	0		

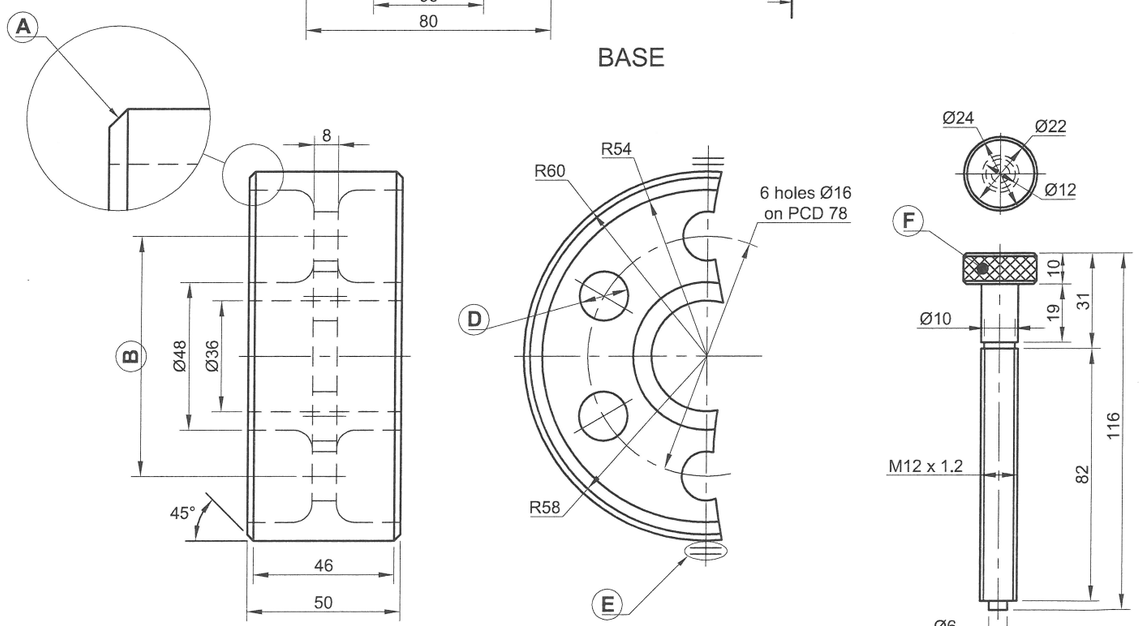
FINAL CONVERTED MARK	CHECKED BY
100	

COMPLETE THE FOLLOWING:
NAME
NAME
SCHOOL
SCHOOL

STAPLE



BASE



PULLEY BOLT

QUESTION 1: ANALYTICAL (MECHANICAL)

Given:
Three parts of a pulley tensioner with a title block and a table of questions.

Instructions:
Complete the table below by neatly answering the questions, which all refer to the accompanying drawings and title block.
Show all calculations. **[30]**

	QUESTIONS	ANSWERS		
1	On what date was the drawing first checked?		1	
2	From what material is the pulley manufactured?		1	
3	Which drawing method was used to create these drawings?		1	
4	How many threaded holes are there in the BASE?		1	
5	How many sets of drawings are there?		1	
6	What is the tolerance allowed on the dimensions?		1	
7	How deep must the M12 thread be?		1	
8	What is feature A called?		1	
9	Determine the dimension at B?		2	
10	What is feature C called?		1	
11	What is the reason for using feature C?		2	
12	What is the size of the hole marked D?		1	
13	What does the double lines at E indicate?		1	
14	What is feature F called?		1	
15	What is the reason for using feature F?		2	
16	Determine the dimension at G.		2	
17	What type of section is shown at SECTION BB?		1	
18	What type of section would cutting plane AA generate?		1	
19	What do the letters PCD stand for?		1	
20	How many Ø16 holes need to be drilled in the PULLEY?		1	
21	Why did they only draw half of the PULLEY?		2	
22	In the box below (ANSWER 22), draw, in neat freehand, the symbol for the projection system used.		4	
	TOTAL		**30**	

ANSWER 22

SYMBOL

ALL DIMENSIONS ARE IN MILLIMETRES.	DATE	CHANGED BY	REVISION DESCRIPTION №
UNLESS OTHERWISE SPECIFIED, TOLERANCES ON DIMENSIONS ARE ± 0,35.	DRAWN BY: Keith	DRAWING SET NO. 2 OF 4	MATERIAL:
	DATE: 15/02/2016	FILE NAME: BASE-53-2016	HEAT TREATMENT: NONE
ALL UNSPECIFIED RADII ARE R6.	CHECKED BY: Ann	MICRO STEEL MANUFACTURING	SUTTON ROAD SYDENHAM 6001 www.microsteel.co.za
	DATE: 18/03/2016		
DRAWING PROGRAM: AUTOCAD 2015	APPROVED BY: Peter	PULLEY TENSIONER	
	DATE: 28/03/2016		
	SCALE: 1 : 2		

PARTS LIST		
PART	QUANTITY	MATERIAL
1. BASE	1	CAST IRON
2. BOLT	1	MILD STEEL
3. PULLEY	1	CAST IRON

NAME	
NAME	2

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S

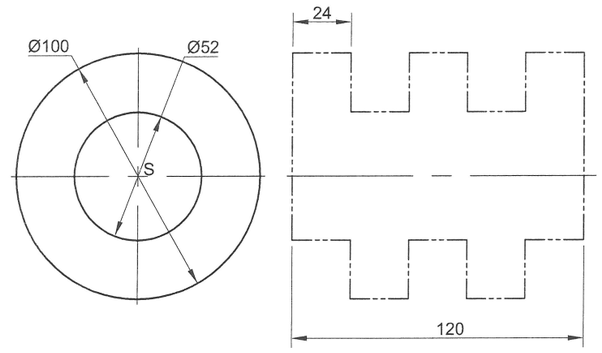
QUESTION 2: LOCI

Given:

- The left view and the profile of a single-start right-hand square thread.
- The position of S on the drawing sheet.

Instructions:

- Draw, to scale 1 : 1, the left view and the front view showing TWO turns of the single-start right-hand square thread according to the given profile.
- Show ALL necessary construction.
- **No hidden detail is required.** **[40]**



ASSESSMENT CRITERIA				
1 GIVEN + CL + CONSTR'	$9\frac{1}{2}$			
2 OUTSIDE HELICES	$12\frac{1}{2}$			
3 INSIDE HELICES	5			
4 SHAFT	6			
5 CURVE QUALITY	7			
TOTAL	**40**			
NAME				
NAME				3

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STAPLE

QUESTION 3: ISOMETRIC DRAWING

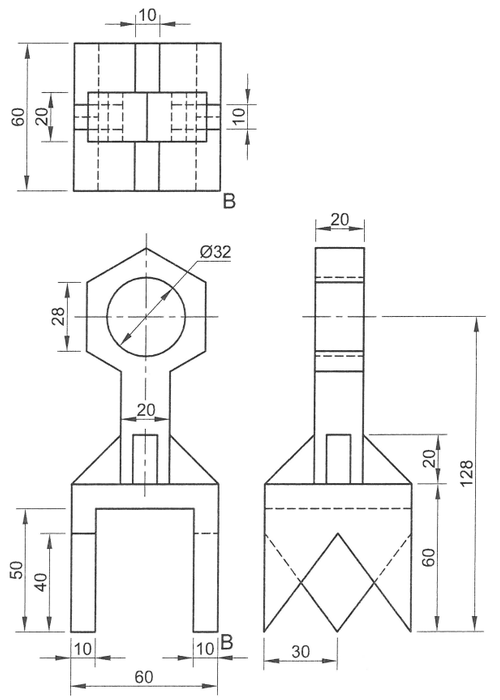
Given:

- The front view, top view and right view of a planter.
- The position of point B on the drawing sheet.

Instructions:

Convert the orthographic views of the planter into a scale 1 : 1 isometric drawing.

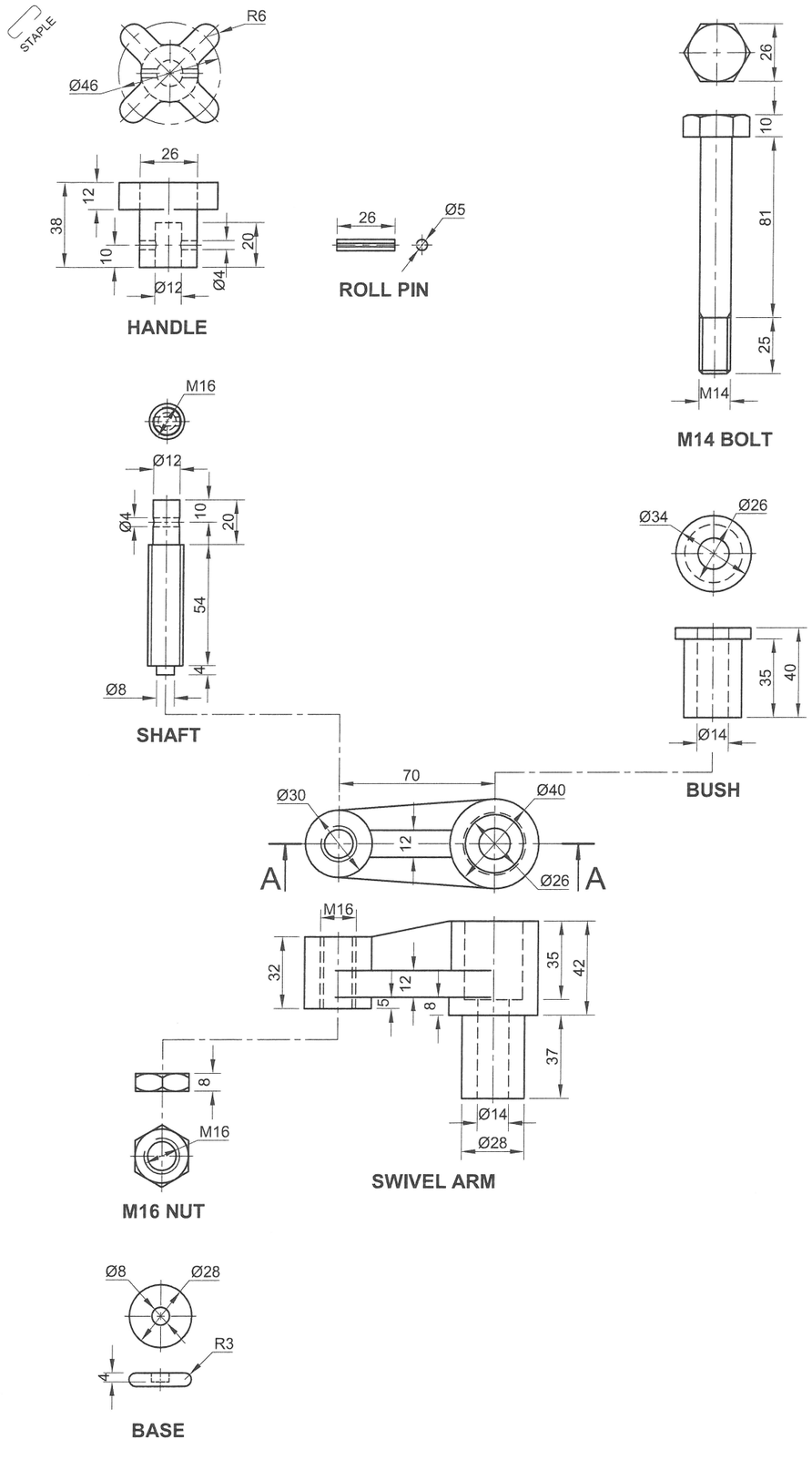
- Make corner B the lowest point of the drawing.
- Show ALL necessary circle and other construction.
- NO hidden detail is required. **[40]**

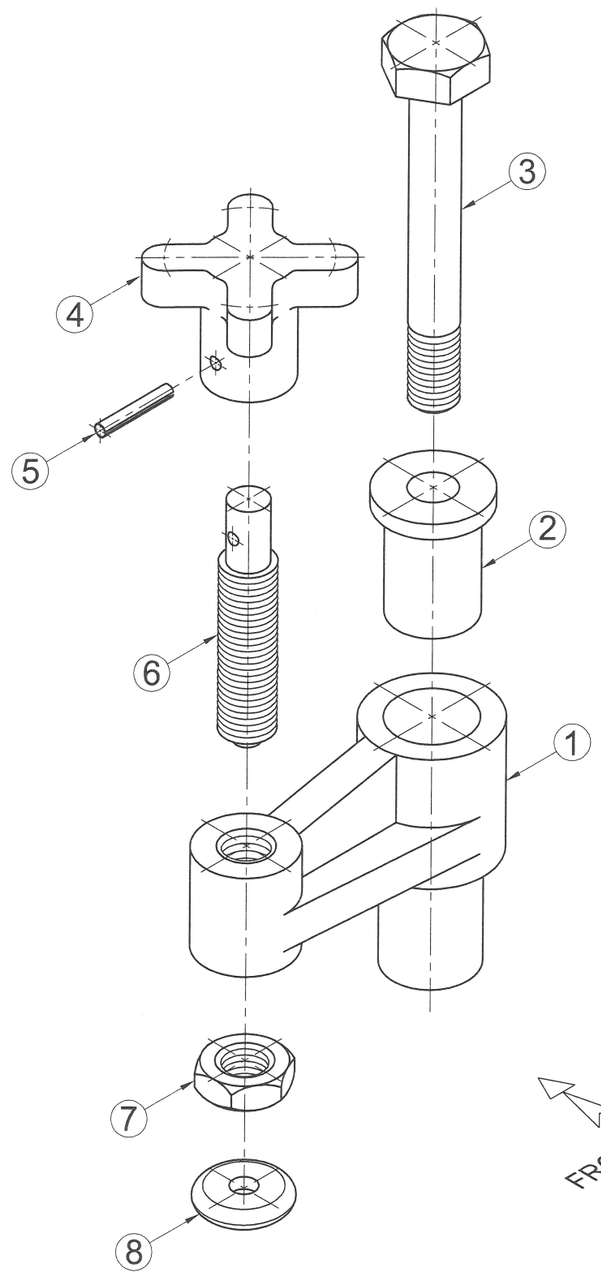


B

ASSESSMENT CRITERIA					
1. AUX. VIEW + PLACING	3				
2. ISO' LINES	13½				
3. NON ISO' LINES	7				
4. HEXAGON	9½				
5. ISO' CIRCLES + CL	7				
TOTAL	**40**				
NAME					
NAME					4

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FRONT VIEW

QUESTION 4: MECHANICAL ASSEMBLY

Given:

- The exploded isometric drawing of the parts of a swivel clamp assembly, showing the position of each part relative to all the others
- Orthographic views of each of the parts of the swivel clamp assembly

Instructions:

- Answer this question on page 6.
- Draw, to scale 1 : 1 and in third-angle orthographic projection, the following view of the assembled parts of the swivel clamp assembly:
  - **4.1 A sectional front view**, on cutting plane A-A, as seen from the direction of the arrow shown on the exploded isometric drawing. The cutting plane is shown on the top view of the swivel arm.
  - **4.2 A top view** of the assembly.
- ALL drawings must comply with the guidelines contained in the *SABS 0111.*

NOTE:

- Show three sides of the bolt head and two sides of the M16 nut in the sectional front view.
- Show all the constructions for the nut.
- No hidden detail is required. **[90]**

PARTS LIST		
PART	QUANTITY	MATERIAL
1. SWIVEL ARM	1	CAST IRON
2. BUSH	1	PLASTIC
3. BOLT	1	MILD STEEL
4. HANDLE	1	CAST IRON
5. ROLL PIN	1	SPRING STEEL
6. SHAFT	1	MILD STEEL
7. M16 NUT	1	MILD STEEL
8. BASE	1	PLASTIC

ALL DIMENSIONS ARE IN MILLIMETRES.	DRAWN BY: JS	CLAMP MANUFACTURING	BUFFALO STREET MTHATHA 5100 www.clamp.co.za
	DATE: 20/01/2016		
ALL UNSPECIFIED RADII ARE R3.	CHECKED BY: AC		
	DATE: 30/02/2016		
DRAWING PROGRAM: CAD 2016	APPROVED BY: GW	TITLE SWIVEL CLAMP	EASTERN CAPE DEPARTMENT BASIC EDUCATION GRADE 11 NOVEMBER 2016
	DATE: 26/03/2016		

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Please turn over

STAPLE

ASSESSMENT CRITERIA				
TOP VIEW				
1. SWIVEL ARM	5			
2. BUSH	1			
3. BOLT	4			
4. HANDLE	$13\frac{1}{2}$			
5. CENTERLINES	6			
ASSEMBLY	7			
SUB TOTAL	**$36\frac{1}{2}$**			

ASSESSMENT CRITERIA				
SECTIONAL FRONT VIEW				
1. SWIVEL ARM	$13\frac{1}{2}$			
2. BUSH	$4\frac{1}{2}$			
3. BOLT	$12\frac{1}{2}$			
4. HANDLE	6			
5. CIRCLIP	$1\frac{1}{2}$			
6. SHAFT	$7\frac{1}{2}$			
7. M16 NUT	$4\frac{1}{2}$			
8. BASE	2			
9. CENTERLINES	$1\frac{1}{2}$			
SUB TOTAL	**$53\frac{1}{2}$**			
TOTAL	**90**			

NAME	
NAME	6

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